Task

Imagine a rabbit population living in an area of grassland. What factors will affect the size of the population?

Cut out the statements.

Discuss with your partner why and how each factor could affect the rabbit population.

Sort them into two piles –

- factors that would increase the rabbit population
- factors that would decrease the rabbit population

Are there any statements that you thought could be put into either pile?

buzzard moves into area	**grassland is built on by developers**	**fox population moves away**
sheep introduced to grassland	**new bacterial disease affects foxes**	**large area of grassland is used to farm salad crops**
grass cover scorched by high temperatures	**increase in number of other small herbivores**	**deadly myxomatosis virus enters rabbit population**
new female rabbits enter area	**grassland ploughed**	**decrease in stoat numbers**
increase in plant that is toxic to rabbits	**overgrazing of grassland by rabbits**	**rabbit population develops resistance to myxomatosis virus**

Teaching notes and answers

increase in rabbit population	decrease in rabbit population
fox population moves away	buzzard moves into area
new bacterial disease affects foxes	grassland is built on by developers
large area of grassland is used to farm salad crops	sheep introduced to grassland
new female rabbits enter area	grass cover scorched by high temperatures
decrease in stoat numbers	myxomatosis virus enters rabbit population
rabbit population develops resistance to myxomatosis virus	grassland ploughed
	increase in plant that is toxic to rabbits
	overgrazing of grassland by rabbits

‘Increase in number of other small herbivores’ – this could go into either group. Ask students to justify their decision. An increase in other small herbivores could lead to increased competition for food. It could lead to an increase in rabbit population due to more prey being available to their predators.

It might be useful to get students to research the habitat and feeding relationships of rabbits before using this activity. Students could use the information to build food chains and a food web for a grassland habitat.

BBC wildlife has some interesting clips to show students about the ecology of rabbits, particularly relevant are the clips ‘Rabbit hunt’ and ‘Boom and bust’ http://www.bbc.co.uk/nature/life/European_Rabbit (the link was available at the time of publishing.